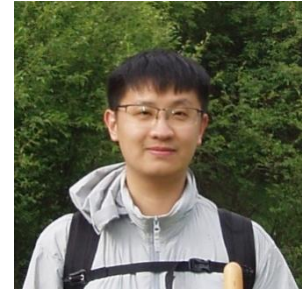


Curriculum vitae (September 2026)

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Research Interests

Wetland Hydrodynamics, Soil Carbon Cycle, Global Change Ecology, Restoration Ecology, Climate Change Impacts on Ecosystems, Digital Soil Mapping, Machine Learning, Remote Sensing Monitoring of Surface Conditions, Hydrodynamic Modelling

Education

2024– Ph.D. Candidate in Hydrodynamics
Nanjing Hydraulic Research Institute, Center for Eco-Environmental Research, Nanjing, China

2020–2023 Joint M.S. in Pedometrics
Institute of Soil Science, Chinese Academy of Sciences, Nanjing, China

2020–2023 M.S. in Forest Ecology
Hunan Agriculture University, Collage of Forestry, Zhengzhou, China

2016–2020 B.S. in Landscape Planning
Hunan Agriculture University, Collage of Landscape Architecture, Zhengzhou, China

Academic and Professional Career

02/2025– Doctoral Research Assistant (NSFC research projects U25A20359 and 52121006)
Department Ecohydraulic Research, Center for Eco-Environmental Research, **Nanjing Hydraulic Research Institute**

2024–2025 Doctoral Research Assistant (NKR&D research project 2023YFC3208902)
Department Water Environmental Informatics, Center for Eco-Environmental Research, **Nanjing Hydraulic Research Institute**

2023–2024 Research Assistant (NKR&D research project 2022YFB3903302)
Department of Earth System Science, School of Resources and Environmental Sciences, **Wuhan University**

2020–2023 Joint Research Assistant (NSFC research project 41930754; NKR&D research projects 2018YFC1801806 and 2018YFC1800104-04; CAS research project XDA28010102)
Soil Resources and Information Research Department, **Institute of Soil Science, Chinese Academy of Sciences**

Awards / Distinctions / Fellowships

2026	The 1 st Prize for Scientific and Technological Progress of Nanjing Hydraulic Research Institute
2025–2027	CAST Young Scientific and Technological Talent Cultivation Program (Doctoral Track)

Memberships

2025–	Student member, Chinese Society for Environmental Sciences (S290000278A)
2025–	Student member, Chinese Society of Natural Resources (S300005247A)
2025–	Student member, Chinese Hydraulic Engineering Society (E081056595A)
2024–	Student member, Ecological Society of Chinese (S280010545S)
2022–	Student member, Soil Science Society of Chinese (C031602678A)

Presentations of Scientific Conference

10/2026	Sun, Z. , Feng, T., Chen, Q. Mineral capacity and ecosystem carbon supply jointly constrain carbon storage in global wetland soils. The 3 rd International Conference on All Material Fluxed in River Ecosystems (AMFR 2026), Beijing, China
05/2026	Sun, Z. , Feng, T., Chen, Q. Enhanced soil carbon sequestration capacity is facilitated by vegetation water use efficiency on Earth's Third Pole. The 6 th Hydro90 Young Hydrologists Academic Symposium, Network
04/2026	Sun, Z. , Feng, T., Chen, Q. Plant input limitation weakens wetland soil carbon sequestration and hence the projected climate mitigation. The 2 nd Watershed Carbon Neutrality Conference, Guangzhou, China
08/2025	Sun, Z. , Chen, Q., Zhang, J. Enhanced soil carbon sequestration capacity is facilitated by vegetation water use efficiency on Earth's Third Pole. The Soil Classification and Soil Geography Professional Committee and the Soil Remote Sensing and Information Professional Committee of the Soil Science Society of China joint academic symposium in 2025, Shenyang, China

Fundings

2026–	Postgraduate Research & Practice Innovation Program of Jiangsu Province (26CXJH7520), CNY60,000
2025–	Postgraduate Thesis Fund of Nanjing Hydraulic Research Institute (Yy626003), CNY60,000

Teaching

2023–2024	Soil- and Bio-geography; Soil Resource Investigation and Evaluation (to undergraduate students), School of Resources and Environmental Sciences, Wuhan University, China
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Journal Services

2025–	Editorial assistant for <i>Research</i>
Peer reviewer	<i>Geoderma</i> , <i>Catena</i> , <i>Journal of Geophysical Research-Biogeosciences</i> , <i>Land Degradation & Development</i> , <i>Frontiers in Plant Science</i> , <i>Environmental Monitoring and Assessment</i> , <i>Water Resources and</i>

Hydropower Engineering (in Chinese), *Chinese Journal of Soil Science* (in Chinese)

Full Publication List by Zheng Sun (without Conference Abstracts)

2026

- Sun, Z.**, Chen, C., Chen, Q., Zhang, J., Zhang, X., Liu, Y., Wang, S., Feng, T., Yan, X., & Qiao, R. (2026). Enhanced soil carbon sequestration capacity is facilitated by vegetation water use efficiency on Earth's Third Pole. *The Innovation Geoscience*, 4(1), 100181. <https://doi.org/10.59717/j.xinn-geo.2026.100181>
- Sun, Z.**, Feng, T., Niu, L., Shi, X., & Chen, Q. (2026). A review on estimation and regulation of sensitive ecological flow processes. *Journal of Hydraulic Engineering*. (in Chinese with English abstract). <https://doi.org/10.3724/j.slxb.20250627>
- Chen, C., Liu, Y., **Sun, Z.**, Chen, Q., Zhang, J., Ma, J., & He, M. (2026). Future risks of cyanobacterial blooms in lakes unveiled by open access data and integrated machine learning models. *Water Research*, 293, 125428. <https://doi.org/10.1016/j.watres.2026.125428>
- Wei, Y., Chen, Y., Wang, J., **Sun, Z.**, Wang, B., Zhang, C., Wu, C., Zhao, T., & Shen, H. (2026). Enhancing farmland soil carbon mapping: Integrating radar backscatter model and machine learning. *Soil & Tillage Research*, 258, 107055. <https://doi.org/10.1016/j.still.2025.107055>
- Wei, L., Wang, J., Wei, Y., **Sun, Z.**, Ma, L., Hong, Y., & Chen, Y. (2026). An innovative mapping framework for soil erodibility integrating spatial association dimensions and machine learning. *Soil & Tillage Research*, 260, 107135. <https://doi.org/10.1016/j.still.2026.107135>
- Wen, H.Y., **Sun, Z.**, Wu, H.Y., Hu, Y.M., & Zhang, G.L. (2026). Magnitude and causes of three decades of subtropical soil acidification after land use change. *Land Degradation & Development*, 37, 2255-2266. <https://doi.org/10.1002/ldr.70243>
- Wei, Y., Chen, Y., **Sun, Z.**, Wang, J., Yu, P., Bai, S., Gu, S., & Liu, Y. (2026). Spatial variation and driving relationship analysis of soil total nitrogen in plateau lake watershed: A case of Qilu Lake Basin in Yunnan. *Geographical Science*, 46(4), 902-915. (in Chinese with English abstract). <https://doi.org/10.13249/j.cnki.sgs.20240953>

2025

- Sun, Z.**, Liu, F., Yang, F., Wang, D., & Zhang, G.L. (2025). Spatial distributions, driving factors, and threshold effects of soil organic carbon stocks in the Tibetan Plateau. *Soil & Tillage Research*, 248, 106457. <https://doi.org/10.1016/j.still.2025.106457>
- Wen, H., Yang, F., **Sun, Z.**, Miao, Z., Hu, J., & Zhang, G. (2025). Asymmetric responses of soil organic carbon stability to shifting dominance of pH-mediated metal-bound organic carbon. *Communications Earth & Environment*, 6, 574. <https://doi.org/10.1038/s43247-025-02565-x>
- Qu, L., **Sun, Z.**, Tian, Z., Schoorl, J.M., Ma, R., Zhao, Y., Wu, Y., & Liang, Y. (2025). Climate-driven spatiotemporal variation of soil erodibility in China. *Environmental Impact Assessment Review*, 115, 107998. <https://doi.org/10.1016/j.eiar.2025.107998>
- Wen, H., **Sun, Z.**, Yang, F., & Zhang, G.L. (2025). Aridity regulates the vital drivers of soil organic carbon content in the Northeast China. *Catena*, 257, 109192. <https://doi.org/10.1016/j.catena.2025.109192>
- Liu, Y., Chen, C., Chen, Q., Zhang, J., **Sun, Z.**, Yan, X., & Huang, Q. (2025). Improve Carbon Budget Assessment in Floodplain Wetlands Using Hydrodynamic and Integrated Machine Learning Models. *Water Resources Research*, 61(10), e2025WR040128. <https://doi.org/10.1029/2025WR040128>

2024

- Sun, Z.**, Liu, F., Wu, H., & Zhang, G.L. (2024). Developing a national black soil map of China through machine learning classification. *Catena*, 240, 107993. <https://doi.org/10.1016/j.catena.2024.107993>
- Wang, J., Wei, Y., **Sun, Z.**, Gu, S., Bai, S., Chen, J., Chen, J., Hong, Y., & Chen, Y. (2024). Optimal Mapping of Soil Erodibility in a Plateau Lake Watershed: Empirical Models Empowered by Machine Learning. *Remote Sensing*, 16, 3017. <https://doi.org/10.3390/rs16163017>
- Song, M., Yang, S., **Sun, Z.**, Zeng, R., Jiang, Z., Song, X., & Zhang, G. (2024). Identification of Mollic Epipedon Using Random Forest and Digital Image Processing Methods. *Chinese Journal of Soil Science*, 55(6), 1501-1511. (in Chinese with English abstract). <https://doi.org/10.19336/j.cnki.trtb.2023121302>

2023

- Sun, Z.**, Liu, F., Wang, D., Wu, H., & Zhang, G. (2023). Improving 3D Digital Soil Mapping Based on Spatialized Lab Soil Spectral Information. *Remote Sensing*, 15, 5228. <https://doi.org/10.3390/rs15215228>
- Sun, Z.**, Cao, Y., Liu, F., Wang, D., Zhang, G., & Wu, H. (2023). Classification and Human Health Risk Prediction of in-situ Leaching Site of Ionic Rare Earth Ore. *Chinese Journal of Soil Science*, 54(5), 1186-1195. (in Chinese with English abstract) . <https://doi.org/10.19336/j.cnki.trtb.2023010301>
- Sun, Z.**, Wu, H., Wang, D., Li, D., & Liu, F. (2023). Scale Effects of Soil Fertility Evaluation on Tobacco Fields in Yunnan-Guizhou Plateau, Southwest China. *Chinese Journal of Soil Science*, 54(4), 800-811. (in Chinese with English abstract). <https://doi.org/10.19336/j.cnki.trtb.2022071002>
- Sun, Z.**, Cao, Y., Wang, D., Liu, F., Song, X., Zhang, G., & Wu, H. (2023). Spatial-temporal evolution characteristics and trend prediction of electroplating sites in the Beijing-Tianjin-Hebei region over the past 30 years. *Ecology and Environmental Sciences*, 32(1), 183-194. (in Chinese with English abstract). <https://doi.org/10.16258/j.cnki.1674-5906.2023.01.020>
- Wu, W., Wang, D., Chen, Z., Liu, F., Long, H., Ma, C., Zhao, Y., Zhang, J., Chen, S., Lu, C., Xu, A., Pei, J., Xu, X., Lei, Q., Huang, Q., Lu, M., Guo, L., Li, Y., **Sun, Z.**, Ruan, Z., Gao, M., Qian, J., Yu, Q., & Wang, X. (2023). Technical Specifications for the Production of Work Baselines and Sampling Point Layout in the 3rd National Soil Survey (Revised Edition), in: The Office of the 3rd National Soil Survey Leading Group of the State Council (Eds.), *Technical Regulations and Specifications for the 3rd National Soil Survey (Revised Edition)*. China Agricultural Press, Beijing, pp. 67-81.
- Zhao, Y., Liu, F., Zhang, G., Sun, X., Chen, J., Wang, L., Yang, L., Pan, J., Sun, F., Zhang, L., Guo, L., Xu, S., Song, X., Wu, H., **Sun, Z.**, Zhao, Y., Yang, R., Huang, W., Pei, J., Gao, M., Yuan, Q., Zhao, Y., Zhu, A., Shi, Z., Wu, K., Li, W., Pan, X., Teng, Y., & Shen, R. (2023). Technical Specifications for Compilation of Soil Type Maps in the 3rd National Soil Survey (Revised Edition), in: The Office of the 3rd National Soil Survey Leading Group of the State Council (Eds.), *Technical Regulations and Specifications for the 3rd National Soil Survey (Revised Edition)*. China Agricultural Press, Beijing, pp. 341-367.

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- Angelini, M.E. (2022). Chapter 2. Global distribution and characteristics of black soils, in: FAO (Eds.). *Global status of black soils*. Rome, pp. 14-70.